Luke Floden

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Engineer who builds and operates across the full stack, with depth in both technical architecture and product execution. Known for high-trust ownership: from platform design and cross-team adoption through production incidents and customer-facing delivery in high-stakes startup environments.

Work Experience

Edapt Schools

Principal Engineer — Aug 2025 - Present

- **Owned delivery of Edapt's multiplayer real-time editor**, enabling section-safe concurrent editing, computed budget fields, and human/AI-agent collaboration across 200-page compliance documents (*Next.js App Router*, *tRPC*, *SSE*).
- **Led a 5-engineer team to take the platform live with 20 California counties**, designing the org model, auth foundations, and multi-org hierarchy — enabling role-based access across county, district, and school relationships so compliance reviewers can audit documents across their entire org tree.
- **Designed the object versioning and history system** to handle concurrent edits across users, services, and AI agents — making every operation attributable and prior states recoverable, sustaining performance across documents with 8,000+ tracked operations.
- **Replaced a fire-and-forget Redis job pattern with a durable job architecture** — atomic claim-and-enqueue, explicit ownership locks, and crash-safe consumer groups — cutting job timeout feedback from 10 minutes to 90 seconds and enabling reconnect-safe status across 3 coordinated services.
- **Re-architected the real-time editor** to bring large-document interactions from over 500ms to under 100ms using optimistic UI, local state caching, and batched network updates, restoring responsive typing and edit feedback.
- **Led recovery from a complete production data loss incident** — replayed 5,518 *Redis* stream events to fully restore 6 days of lost customer data within 2 hours — then drove post-incident remediation and added *Sentry* monitoring and safeguards to prevent recurrence.
- **Established a queryable wide-logging standard across services**, propagating trace IDs through durable jobs to speed root-cause analysis for incidents and data-loss bugs. (*OpenTelemetry*)

Cadence Onefive

Lead Software Engineer — Jun 2024 - Jan 2025

- **Led a distributed engineering team across 5 time-zones**, taking Momentum™ (a SaaS accelerating building retrofits for decarbonization) from overdue and fragmented to stable, fast, and design-driven.
- **Cut weeks off delivery** upgrading a legacy monolith to *Laravel*/*Livewire*, overhauling component lifecycle management, and moving the build chain to *Vite*; saving 5+ dev hours per week and halving code review surface.
- **Established weekly all-stakeholder demos and review loops**, restoring alignment and reducing delivery drift on an overdue product.
- **Served as the bridge between design and engineering** during a key team transition, partnering with the CEO on designer hiring and helping move the org to a unified, *Figma*-driven design system.

Senior Software Engineer — Feb 2024 - Jun 2024

- **Delivered a complete UI overhaul to maximize accessibility** (semantic *HTML*, keyboard navigation) and mobile support. Introduced a custom component library; immediately reducing copy-paste debt and increasing feature velocity.
- **Launched automated linting and formatting across the stack**; surfaced and fixed long-standing UI bugs and broken markup, improving reliability for critical real estate, finance, and utility stakeholders.

Curai Health

Full Stack Engineer III — Sep 2022 - Jun 2023

- **Shipped a cross-platform app to 20k+ users** providing users access to a dedicated care team for real-time, personalized medical support. Built with *React Native*, *TanStack Query*, and *Redux*.
- **Architected an Enterprise-level SDK and API** enabling third-party integrations without disrupting the core product. Written in *Typescript*, this *NPM* package included custom routing and cross-document messaging.
- **Cut load times in half** and redundant data fetches through a full migration from *Redux* to *TanStack Query*, giving our web app a native feel.
- **Reduced user churn rates through A/B experimentation** during the development of a new onboarding and paywall flow.

Full Stack Engineer II — Dec 2021 - Sep 2022

- **Drove cross-team adoption of a new external gateway service** across four dev teams through a staged monolith-to-microservices rollout, establishing type-safe service contracts and automatic API documentation as the new platform standard. (*FastAPI*)
- **Built a self-service partner data intake portal** that auto-detected column mappings across Excel and CSV uploads, surfaced field mismatches for in-app resolution, and eliminated the monthly back-and-forth of corrected file exchanges with enterprise partners.
- **Instituted automated documentation and strict type safety** in *FastAPI* services, improving transparency and reliability across the backend.

ML Engineering Intern — Sep 2021 - Dec 2021

- **Built an interactive ML pipeline analysis tool** (*React* + *dc.js*), integrating early-bias detection and safeguards across user demographics.
- **Ran structured user studies** to inform design decisions, closing the loop between engineering, product, and design.

University of Louisville

Research/Teaching Assistant — Jun 2017 - May 2021

- **Guided a course on Operating Systems** under Dr. Nihat Altiparmak, instructing students through core OS concepts, *C* programming, and systems-level debugging.
- **Provided direct technical mentorship**, helping students with 1-on-1 sessions that often involved over an hour of pair programming.

Justice Av Solutions

Software Engineer — Jan 2018 - Aug 2019

- **Built and debugged .NET desktop apps and Vue.js frontends** for court recording systems serving 17 countries.
- **Integrated TCP/IP networking tools** and low-level troubleshooting for robust device-driver performance in mission-critical courtroom AV environments.

Education

University of Louisville

Master of Computer Science Engineering	2020 - 2021
Bachelor of Computer Engineering & Computer Science	2016 - 2020

Skills

React, Next.js	*Svelte (Kit)*	*TypeScript, tRPC*	*Node.js, Bun, Vite*
Tailwind	*TanStack Query, Redux*	*Laravel, Livewire*	*Redis, PostgreSQL*
FastAPI, Flask	*AWS*	*Figma*	*Jest, Cypress, Playwright*
Docker, Sentry	*WorkOS*		